The empirical recurrence for OEIS A264295, proved by transfer matrix

Adrian Perez Fontelles
Independent researcher

1 September 2026

Abstract

OEIS A264295 counts the arrangements of $0, 1, \ldots, LW-1$ in a grid of $W = 4$ rows in which every value moves from its home cell by an index change on a short list, and it carries an empirical recurrence of order 21 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. Such an arrangement is a perfect matching between cells and values, and because a value's row moves by at most $R = 1$, the matching splits into independent decisions row by row: after the cells of rows $0, \ldots, i$ are filled, every value whose home row is at most $i - R$ has been used, and only $2R$ home rows are partly used. Carrying those as bitmasks makes the count a walk on $S = 256$ vertices, so it is C-finite and the conjectured recurrence is decided by a finite exact computation.

2020 Mathematics Subject Classification. 05A05, 05A15, 15A15.

1 The conjecture

OEIS A264295 is "Number of (n+1)X(3+1) arrays of permutations of 0..n*4+3 with each element having directed index change 0,1 0,-2 1,0 or -1,0.". It has offset 1 and begins

$$5, 8, 89, 192, 1413, 4352, 24781, 90112, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 5*a(n-1) +2*a(n-2) -34*a(n-3) +79*a(n-4) -103*a(n-5) +60*a(n-6) -220*a(n-7) +610*a(n-8) -142*a(n-9) -1276*a(n-10) +1276*a(n-11) +142*a(n-12) -610*a(n-13) +220*a(n-14) -60*a(n-15) +103*a(n-16) -79*a(n-17) +34*a(n-18) -2*a(n-19) -5*a(n-20) +a(n-21)

As of the "Last modified" line on the live entry (Nov 11 2015 07:36:36, revision 4) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The object is a matching

The grid has $W = 4$ rows across and holds each of $0, 1, \ldots, LW - 1$ exactly once. The value v has a HOME cell, namely its place in the plain row-major filling, and putting v in cell (i, j) changes its index by $(i - h_i,\ j - h_j)$ where (h_i, h_j) is that home cell. The entry allows the index changes

$$(i - h_i,\ j - h_j)\ \in\ \{(-1, 0),\ (0, -2),\ (0, 1),\ (1, 0)\},$$

the list being the entry's own, read with the sign convention its wording uses. An admissible arrangement is therefore exactly a perfect matching in the bipartite graph joining each cell to the values it may hold.

3 The matching is a walk

Let $R = 1$ be the largest $|i - h_i|$ on the list. A value whose home row is h can only be placed in rows $h - R, \dots, h + R$. So once the cells of rows $0, \dots, i$ are filled, every value with home row at most $i - R$ has been used up, and the only home rows that can be partly used are $i - R + 1, \dots, i + R$ — exactly $2R$ of them. Take as the state the tuple of $2R$ bitmasks recording which values of those home rows are already placed. One step fills the cells of one further row, drawing each of its W values from a home row within R and at an allowed column offset, and the home row that closes at that step must come out full. Nonexistent home rows — before the first and after the last — are carried as full, which makes the starting and accepting states the same one.

Hence a is a walk count on $S = 256$ vertices,

$$a(n) \;=\; \iota^{\top} M^{\,n+1} \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms; in particular a is C-finite.

4 The proof

Lemma 1. *Let $q(t) = t^{21} - \sum_i c_i t^{21-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^{\,j} q(M) \tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^j q(M) \tau = 0$ for all large j; and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+21} - \sum_i c_i M^{j+21-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n) \;=\; 5\,a(n-1)+2\,a(n-2)-34\,a(n-3)+79\,a(n-4)-103\,a(n-5)+60\,a(n-6)-220\,a(n-7)+610\,a(n-8)-142\,a($$

for every $n > 19$.

Proof. The vector $w = q(M)\tau$ was formed by 21 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^j w$ evaluated for $j = 0, 1, \dots,$ again exactly; those integers vanish from the index corresponding to $n = 19 + 1$ onwards and the run continued until the working vector was identically zero, settling every larger j at once. □

5 Verification

Three checks, each independent of the algebra above.

First, the reading of the entry's wording was tested against the entries themselves by brute force before any of this was built, and separately for each of the three sign conventions the family uses: "$(\pm, \pm)$ a, b" signs each coordinate independently, "$\pm(.,.)$ a, b" signs the PAIR and may write the second coordinate negative, and "directed a, b" is taken literally. Enumerating all 720 arrangements of a 2×3 grid reproduces the first term of an entry of each convention. Testing only one of them would have proved only that one: reading the second convention as the first silently drops displacements, and thirty-eight entries disagreed with their own published data until the conventions were separated.

Second, the transfer model reproduces all 22 terms this entry publishes, exactly, in integer arithmetic.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A264295.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[3] L. Lovász and M. D. Plummer, *Matching Theory*, AMS Chelsea, 2009.

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.